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Microphone

Abstract

A microphone is provided. The microphone includes a housing having an accommodating cavity, a circuit board fixed to the accommodating cavity, an interface assembly fixed to an outer side of the housing and electrically connected to the circuit board, and a microphone unit fixed to the circuit board. The housing includes a shell having the accommodating cavity and a cover plate fixed to the housing and covering the accommodating cavity, the interface assembly is fixed to one side of the shell away from the cover plate, an insertion direction of the interface assembly is perpendicular to a surface of the circuit board. The microphone cancels the weak connecting plate, which can enhance the connection strength between interface assembly and the housing, which is conducive to miniaturizing the microphone and allowing it to flexibly adapt to different installation methods.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a technical field of microphone technology, in particular to a microphone applied in vehicle entertainment system.

BACKGROUND

(2) Vehicles each generally has a vehicle-mounted entertainment system with functions such as radio, video playback, and karaoke, for the karaoke function, according to habits of people, microphones are an essential device among. Referring to FIG. 1, a microphone typically includes a housing **10** having an accommodating cavity and a circuit board fixed in the accommodating cavity.

(3) In related technologies, in order to achieve interaction between the microphone and the vehicle-mounted entertainment system, an interface assembly **20** is usually disposed on the microphone, a connector of the vehicle-mounted entertainment system is plugged into the interface assembly **20** to achieve connection between the vehicle-mounted entertainment system and the microphone. The interface assembly **20** is fixed on the housing **10** through a connecting plate **30**, and the interface assembly **20** is disposed at one end of the connecting plate **30** away from the circuit board, so that there is a certain distance between the interface assembly **20** and the housing **10**; an insertion direction of the interface assembly **20** is parallel to an extension direction of a surface of the circuit board, which makes the connecting plate **30** prone to breakage due to strong pulling during an

insertion process between the connector and the interface assembly 20, thereby resulting in damage to the microphone.

(4) Therefore, it is necessary to provide a new microphone.

SUMMARY

(5) The present disclosure aims to provide a microphone applied in vehicle entertainment system to solve a problem that a connecting plate of the microphone is easily broken when the microphone is inserted into a connector in related technologies.

(6) The technical solutions of the present disclosure are as follows.

(7) The present disclosure provides a microphone, the microphone includes a housing having an accommodating cavity, a circuit board fixed to the accommodating cavity, an interface assembly fixed to an outer side of the housing and electrically connected to the circuit board, and a microphone unit fixed to the circuit board. The housing includes a shell having the accommodating cavity and a cover plate fixed to the housing and covering the accommodating cavity, the interface assembly is fixed to one side of the shell away from the cover plate, an insertion direction of the interface assembly is perpendicular to a surface of the circuit board, and the shell defines a pickup channel for providing sound signals to the microphone unit.

(8) Optionally, the interface assembly includes an interface socket fixed to the shell and having an inserting cavity, and a contact pin fixed to the inserting cavity, a length extension direction of the inserting cavity is perpendicular to the surface of the circuit board, and the contact pin is electrically connected to the circuit board.

(9) Optionally, two ends of the interface socket are communicated to form the inserting cavity, the interface socket is connected to the shell to be a whole, and the contact pin is fixed to the shell.

(10) Optionally, a component surface is formed at a first side of the circuit board where the microphone unit is disposed, and an acoustic surface is formed at a second side of the circuit board. The cover plate is disposed opposite to the component surface, and the cover plate is fixed to the shell. The pickup channel is disposed opposite to the acoustic surface, and the pickup channel is exposed to the acoustic surface.

(11) Optionally, the cover plate is glued to the shell.

(12) Optionally, the circuit board defines a first through hole communicating the component surface and the acoustic surface, the microphone unit covers a first end of the first through hole located at the component surface, and the pickup channel communicates with a second end of the first through hole located at the acoustic surface.

(13) Optionally, the pickup channel includes a first channel formed by extending inward from a surface of the shell, and a second channel formed by bending from one end of the first channel away from the surface of the shell toward one side close to the acoustic surface and extending to an inner wall of the shell, and the second channel communicates with the first through hole.

(14) Optionally, the microphone further includes a fixing component respectively fixed to the shell and one side of the circuit board having the acoustic surface, the fixing component defines a second through hole, and the second through hole communicates with the first through hole and the second channel.

(15) Optionally, the shell includes a first surface and a second surface, the first surface is adjacent to the second surface, the first surface is disposed opposite to the acoustic surface, the interface assembly is disposed on the first surface, and the first channel forms a pickup hole on the surface of the shell.

(16) The beneficial effects of the present disclosure are as following.

(17) Since the interface assembly is directly fixed on the housing, compared to the related technologies where a connecting plate is set up to fix the interface assembly on the housing, the present disclosure cancels the weak connecting plate, which can enhance the connection strength between interface assembly and the housing, which is conducive to miniaturizing the microphone and allowing it to flexibly adapt to different installation methods; furthermore, the insertion

direction of the interface assembly is perpendicular to the surface of the circuit board, so that there will be no pulling during insertion, which will loosen or even break between the interface assembly and the housing, thereby avoiding damage to the microphone.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic diagram of a microphone in related technologies.
- (2) FIG. 2 is a schematic diagram of a microphone according to one embodiment of the present disclosure.
- (3) FIG. 3 is a schematic diagram of a right-side view of the microphone according to one embodiment of the present disclosure.
- (4) FIG. 4 is a schematic diagram of a cross-sectional view taken along the line A-A shown in FIG. 3.

DETAILED DESCRIPTION

- (5) The following is a further explanation of the present disclosure in conjunction with the accompanying drawings and implementation methods.
- (6) Referring to FIGS. 2 to 4, A microphone includes a housing 1 having an accommodating cavity 111, a circuit board 2 fixed in the accommodating cavity 111, an interface assembly 3 fixed to the outer side of the housing 1 and electrically connected to the circuit board 2, and a microphone unit 5 fixed to the circuit board 2. The housing 1 includes a shell 11 having an accommodating cavity 111 and a cover plate 12 fixed to the shell 11 and covering the accommodating cavity 111, the interface assembly 3 is fixed to one side of the shell 11 away from cover plate 12, the insertion direction of the interface assembly 3 is perpendicular to a surface of the circuit board 2, and the shell 11 defines a pickup channel 4 for providing sound signals to the microphone unit 5.
- (7) Due to the fact that the interface assembly 3 is directly fixed on the housing 1, compared to the related technologies (as shown in FIG. 1) where a connecting plate 30 is set up to fix the interface assembly 20 on the housing 10, the present disclosure cancels the weak connecting plate 30, which can enhance the connection strength between interface assembly 3 and the housing 1, which is conducive to miniaturizing the microphone and allowing it to flexibly adapt to different installation methods; furthermore, the insertion direction of the interface assembly 3 is perpendicular to the surface of the circuit board 2, so that there will be no pulling during insertion, which will loosen or even break between the interface assembly 3 and the housing 1, thereby avoiding damage to the microphone.
- (8) Referring to FIGS. 2 to 4, furthermore, the interface assembly 3 includes an interface socket 31 fixed to the shell 11 and having an inserting cavity 311, and a contact pin 32 fixed to the inserting cavity 311, the length extension direction of the inserting cavity 311 is perpendicular to the surface of the circuit board 2, and the contact pin 32 is electrically connected to the circuit board 2. The connector is inserted into the inserting cavity 311 along the length extension direction of the inserting cavity 311, and the length extension direction of the inserting cavity 311 is perpendicular to the surface of the circuit board 2, allowing the connector to be inserted into the inserting cavity 311 along the direction perpendicular to the surface of the circuit board 2; after inserting the connector into the inserting cavity 311, the connector is connected to the contact pin 32, and the contact pin 32 is electrically connected to the circuit board 2, to achieve electrical connection between the terminal device and the microphone.
- (9) Referring to FIG. 2 and FIG. 4, in one embodiment, two ends of the interface socket 31 are communicated to form the inserting cavity 311, and the interface socket 31 is connected to the shell 11 to be a whole, and the contact pin 32 is fixed to the housing 1, making it easy to shape the microphone.

(10) Referring to FIG. 4, furthermore, a component surface **22** is formed at a first side of the circuit board **2** where the microphone unit **5** is disposed, and an acoustic surface **21** is formed at a second side of the circuit board **2**. The cover plate **12** is disposed opposite to the component surface **22**, and the cover plate **12** is fixed to the shell **11**. A pickup channel **4** is disposed opposite to the acoustic surface **21**, and the pickup channel **4** is exposed to the acoustic surface **21**. Specifically, the cover plate **12** is buckled and fixed to the shell **11**, so that there is a gap at the connection between the cover plate **12** and the shell **11**, from which water droplets or moisture can corrode the circuit board **2**. Due to the sealed connection between the cover plate **12** and the shell **11**, this gap can be sealed to prevent water droplets or moisture from corroding the circuit board **2**, thereby avoiding the microphone from not working due to damage to the circuit board **2**; meanwhile, due to the cover plate **12** is disposed opposite to the component surface **22**, the pickup channel **4** is disposed opposite to the acoustic surface **21**, it will not interfere with the normal use of the pickup channel **4**.

(11) Referring to FIGS. 2 to 4, in one embodiment, the cover plate **12** is glued to the shell **11**, which means that there is a glue component located between the cover plate **12** and the shell **11**, and the connection position between the cover plate **12** and the circuit board **2** is filled with the glue.

(12) Referring to FIG. 4, furthermore, the shell **11** defines the pickup channel **4**, a first end of the pickup channel **4** is exposed on the acoustic surface **21**, and a second end of the pickup channel **4** is exposed outside of the shell **11**, the pickup channel **4** can transmit the sound to achieve normal operation of the microphone.

(13) Referring to FIG. 4, in one embodiment, the circuit board **2** defines a first through hole **23** communicating the component surface **22** and the acoustic surface **21**, the microphone unit **5** covers a first end of the first through hole **23** located at the component surface **22**, and the pickup channel **4** communicates with a second end of the first through hole **23** located at the acoustic surface **21**. The sound signals can enter the microphone through the pickup channel **4** and enter the microphone unit **5** through the first through hole **23**, the microphone unit **5** can convert the sound signals into electrical signals, and the terminal device receives the electrical signals through the connector to achieve the operation of the vehicle-mounted entertainment system.

(14) Referring to FIG. 4, in one embodiment, the pickup channel **4** includes a first channel **41** formed by extending inward from the surface of the shell **11**, and a second channel **42** formed by bending from one end of the first channel **41** away from the surface of the shell **11** toward the side close to the acoustic surface **21** and extending to the inner wall of the shell **11**, and the second channel **42** communicates with the first through hole **23**. The first channel **41** and the second channel **42** are connected in a 7-shaped shape.

(15) Referring to FIG. 4, in one embodiment, the microphone further includes a fixing component **6** fixed to the shell **11** and one side of the circuit board **2** having the acoustic surface **21**, respectively. The fixing component **6** defines a second through hole **61**, the second through hole **61** communicates with the first through hole **23** and the second channel **42**, respectively. Specifically, the fixing component **6** may be a fixing plate, and the fixing component **6** fixes the circuit board **2** to the inner wall of the shell **11**; the fixing component **6** seals one end of the second channel **42** away from the first channel **41**, so that the sound signals can only propagate from the second through hole **61** to the first through hole **23**, thereby ensuring the integrity of the sound signals received by the microphone unit **5**. In one embodiment, a size of the second through hole **61** is equal to a size of the second channel **42**, and the size of the second through hole **61** is greater than a size of the first through hole **23**, which can avoid sound leakage and further ensure the integrity of the sound signals received by the microphone unit **5**.

(16) Referring to FIG. 4, in one embodiment, the shell **11** includes a first surface **112** and a second surface **113**, and the first surface **112** is adjacent to the second surface **113**. The first surface **112** and the acoustic surface **21** are oppositely disposed. The interface assembly **3** is disposed on the

first surface **112**, the first channel **41** forms a pickup hole **411** on the second surface **113** of the shell **11**, thereby providing a certain distance between the interface socket **31** and the pickup hole **411**, after the connector is inserted into the interface socket **31**, and when the microphone is held, it will not interfere with the sound signals received by the pickup hole **411**.

(17) The above description is merely embodiment of the present disclosure, and is not therefore intended to limit patent scopes of the present disclosure, and any equivalent structure or equivalent process transformation made by using the specification and drawings of the present disclosure, or directly or indirectly applied to other related technical fields, is included within the patent scopes of the present disclosure.

Claims

1. A microphone comprising: a housing comprising an accommodating cavity; a circuit board fixed in the accommodating cavity; an interface assembly fixed to an outer side of the housing and electrically connected to the circuit board; and a microphone unit fixed to the circuit board; wherein the housing comprises a shell having the accommodating cavity and a cover plate fixed to the housing and covering the accommodating cavity, the interface assembly is fixed to one side of the shell away from the cover plate, an insertion direction of the interface assembly is perpendicular to a surface of the circuit board, and the shell defines a pickup channel for providing sound signals to the microphone unit; a component surface is formed at a first side of the circuit board where the microphone unit is disposed, and an acoustic surface is formed at a second side of the circuit board; the cover plate is disposed opposite to the component surface, and the cover plate is fixed to the shell; the pickup channel is disposed opposite to the acoustic surface, and the pickup channel is exposed to the acoustic surface; the circuit board defines a first through hole communicating the component surface and the acoustic surface, the microphone unit covers a first end of the first through hole located at the component surface, and the pickup channel communicates with a second end of the first through hole located at the acoustic surface; the pickup channel comprises a first channel formed by extending inward from a surface of the shell, and a second channel formed by bending from one end of the first channel away from the surface of the shell toward one side close to the acoustic surface and extending to an inner wall of the shell, and the second channel communicates with the first through hole.
 2. The microphone according to claim 1, wherein the interface assembly comprises an interface socket fixed to the shell and having an inserting cavity, and a contact pin fixed to the inserting cavity, a length extension direction of the inserting cavity is perpendicular to the surface of the circuit board, and the contact pin is electrically connected to the circuit board.
 3. The microphone according to claim 2, wherein two ends of the interface socket are communicated to form the inserting cavity, the interface socket is connected to the shell to be a whole, and the contact pin is fixed to the shell.
 4. The microphone according to claim 1, wherein the cover plate is glued to the shell.
 5. The microphone according to claim 1, wherein the microphone further comprises a fixing component respectively fixed to the shell and one side of the circuit board having the acoustic surface, the fixing component defines a second through hole, and the second through hole communicates with the first through hole and the second channel.
 6. The microphone according to claim 1, wherein the shell comprises a first surface and a second surface, the first surface is adjacent to the second surface, the first surface is disposed opposite to the acoustic surface, the interface assembly is disposed on the first surface, and the first channel forms a pickup hole on the surface of the shell.
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